

Laboratory Report

Course: Discrete Control Systems
Lab No.: 1 – Modeling of Linear Discrete
Systems

HDU ID: 22320605, 22320630

Name: Cai Xuanhao, Zhu Chenhao

Variant: 22

| Laboratory Report

Course: Discrete Control Systems

Lab No.: 1 – Modeling of Linear Discrete Systems

HDU ID: 22320605, 22320630

Name: Cai Xuanhao, Zhu Chenhao

Variant: 22

| 1. Objective

1. Study the **zero-order hold (ZOH)** and basic techniques for modeling linear discrete systems in **Simulink**.
 2. Investigate the influence of feedback gain (**K_FB**) on system stability and transient response.
 3. Determine:
 - **Neutral stability boundary**
 - **Oscillatory stability boundary**
 - **Time-optimal gain**
-

| 2. Theoretical Background

A **pulse system** consists of a **continuous part (CP)** and a **pulse element (PE)**. The PE performs:

- **Time quantization:** ($t = kT$)
- **Pulse modulation** (amplitude/width)

The **zero-order hold (ZOH)** reconstructs the signal by holding the last sampled value constant until the next sample.

The **z-domain transfer function** of ZOH + integrator is:

$$G(z) = K_{CO} \cdot T \cdot \frac{1}{z-1}$$

Closed-loop characteristic equation (unity feedback):

$$1 + K_{FB} \cdot K_{CO} \cdot T \cdot \frac{1}{z-1} = 0 \quad \Rightarrow \quad z = 1 - K_{FB} \cdot K_{CO} \cdot T$$

Variant 22 parameters:

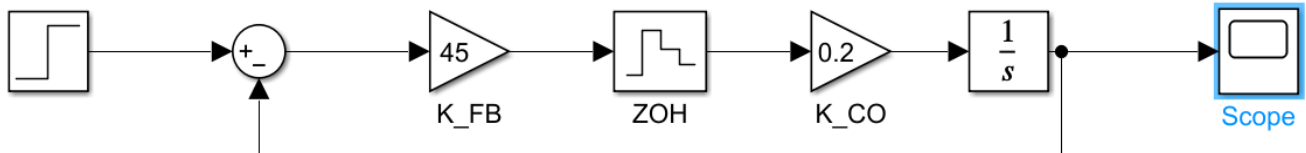
- Sampling period: $T = 0.25$
- Plant gain: $K_{CO} = 0.2$

Thus:

$$z = 1 - 0.05K_{FB}$$

Boundary	Root	Condition	K_FB
Neutral	($z = 0$)	Output $\rightarrow$ constant	20
Oscillatory	($z = -1$)	Sustained square wave	40

3. Simulink Model



4. Experimental Results

4.1 Step Responses for Different K_FB

K_FB	Response Behavior	Steady-State	Figure
16	Fast rise, no overshoot , settles at 1	1	Fig. 1
20	Fastest monotonic rise $\rightarrow$ constant	1	Fig. 2
30	Damped oscillation $\rightarrow$ settles	1	Fig. 3
40	Sustained square-wave oscillation ($0 \leftrightarrow 2$)	—	Fig. 4
45	Divergent oscillation ($\times 10^7$)	—	Fig. 5

4.2 Stability Boundaries

- **Neutral boundary:** ($K_{FB} = 20$) $\rightarrow$ output converges to constant
- **Oscillatory boundary:** ($K_{FB} = 40$) $\rightarrow$ sustained oscillation
- **Unstable region:** ($K_{FB} > 40$)

4.3 Influence of Zero-Order Hold

- **Without ZOH:** pure integrator + proportional feedback $\rightarrow$ **stable for any (K_{FB})**
- **With ZOH:** introduces **sample-and-hold delay** $\rightarrow$ reduces phase margin
- Result: system **becomes unstable** when ($K_{FB} > 40$)

Conclusion: ZOH is the **primary cause of instability** in discrete control systems.

| 4.4 Time-Optimal Control

- ($K_{FB} = 20$):
 - Pole at $z = 0 \rightarrow$ fastest possible response
 - Zero overshoot
 - Optimal performance per theory
-

| 5. Conclusions

1. Successfully modeled a discrete control system with ZOH in Simulink (Variant 22).
 2. Verified theoretical stability boundaries:
 - Neutral: ($K_{FB} = 20$)
 - Oscillatory: ($K_{FB} = 40$)
 3. ZOH significantly degrades stability due to sampling delay.
 4. Higher (K_{FB}) $\rightarrow$ stronger oscillation; maximum sustained oscillation at ($K_{FB} = 40$).
 5. Time-optimal gain: ($K_{FB} = 20$) (pole at origin, fastest settling).
-

| Appendix – Simulation Plots

| Fig. 1 – ($K_{FB} = 16$) (Stable, slower)

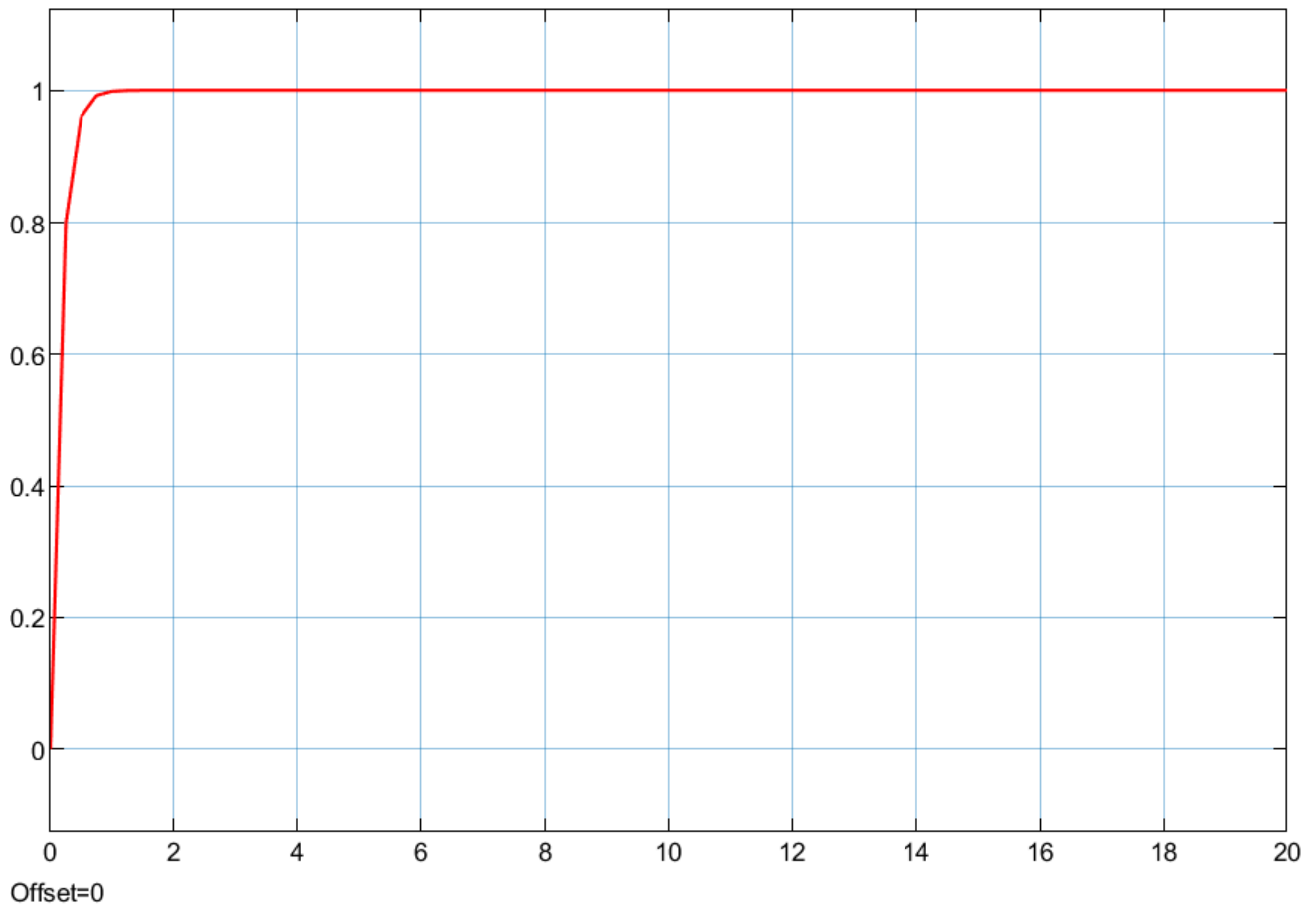


Fig. 2 – ($K_{FB} = 20$) (Time-Optimal)

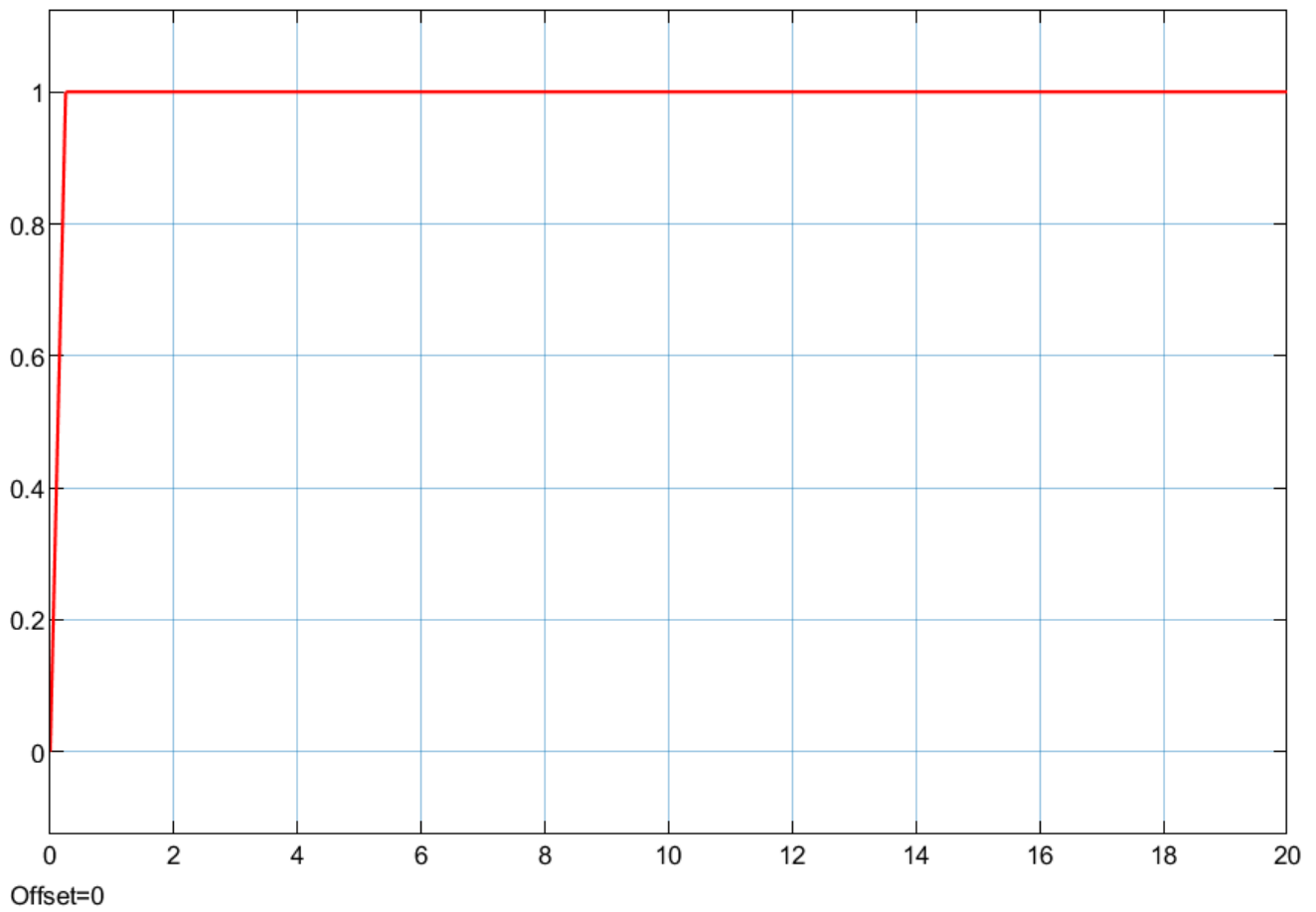


Fig. 3 – ($K_{FB} = 30$) (Damped Oscillation)

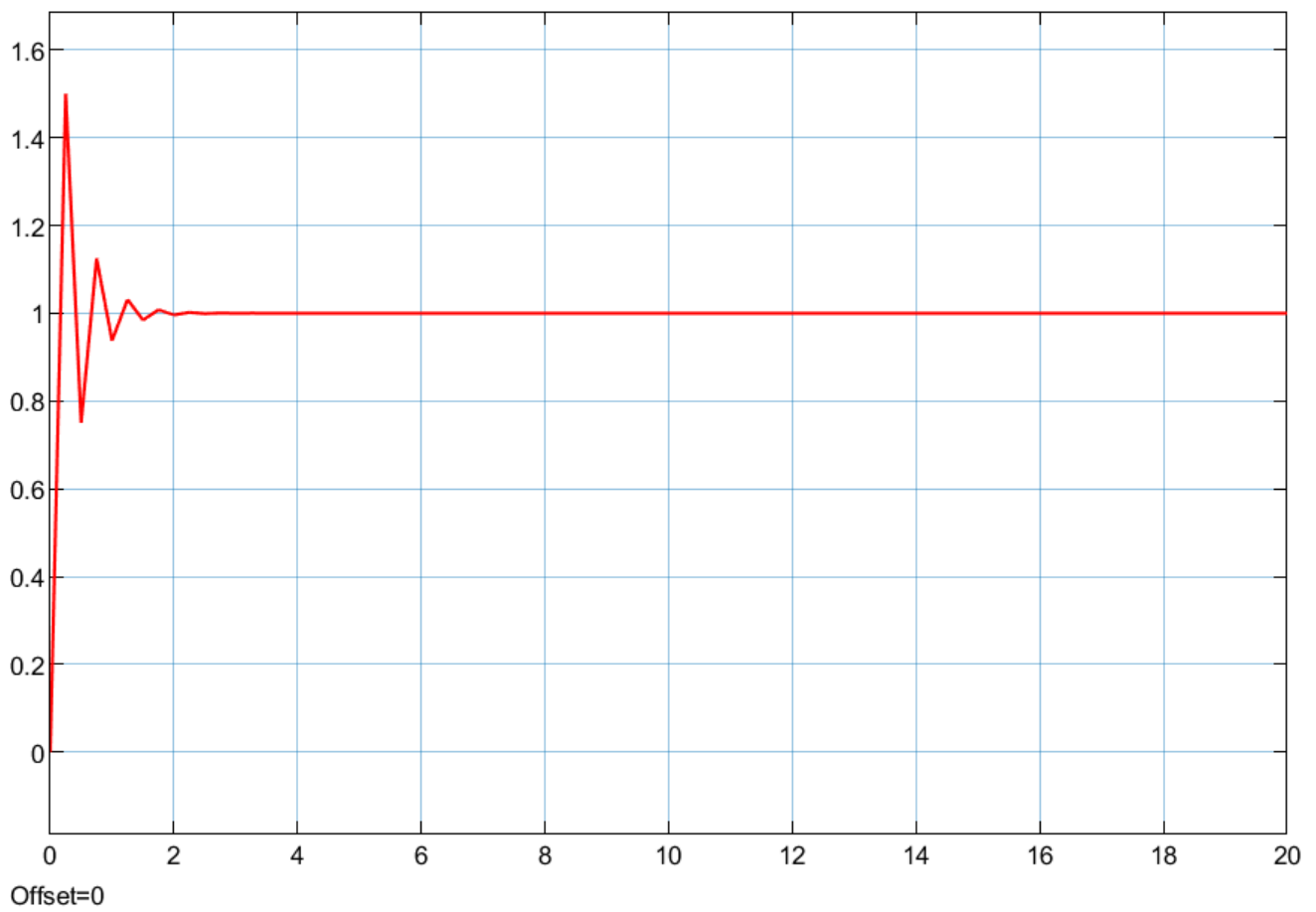


Fig. 4 – ($K_{FB} = 40$) (Oscillatory Boundary)

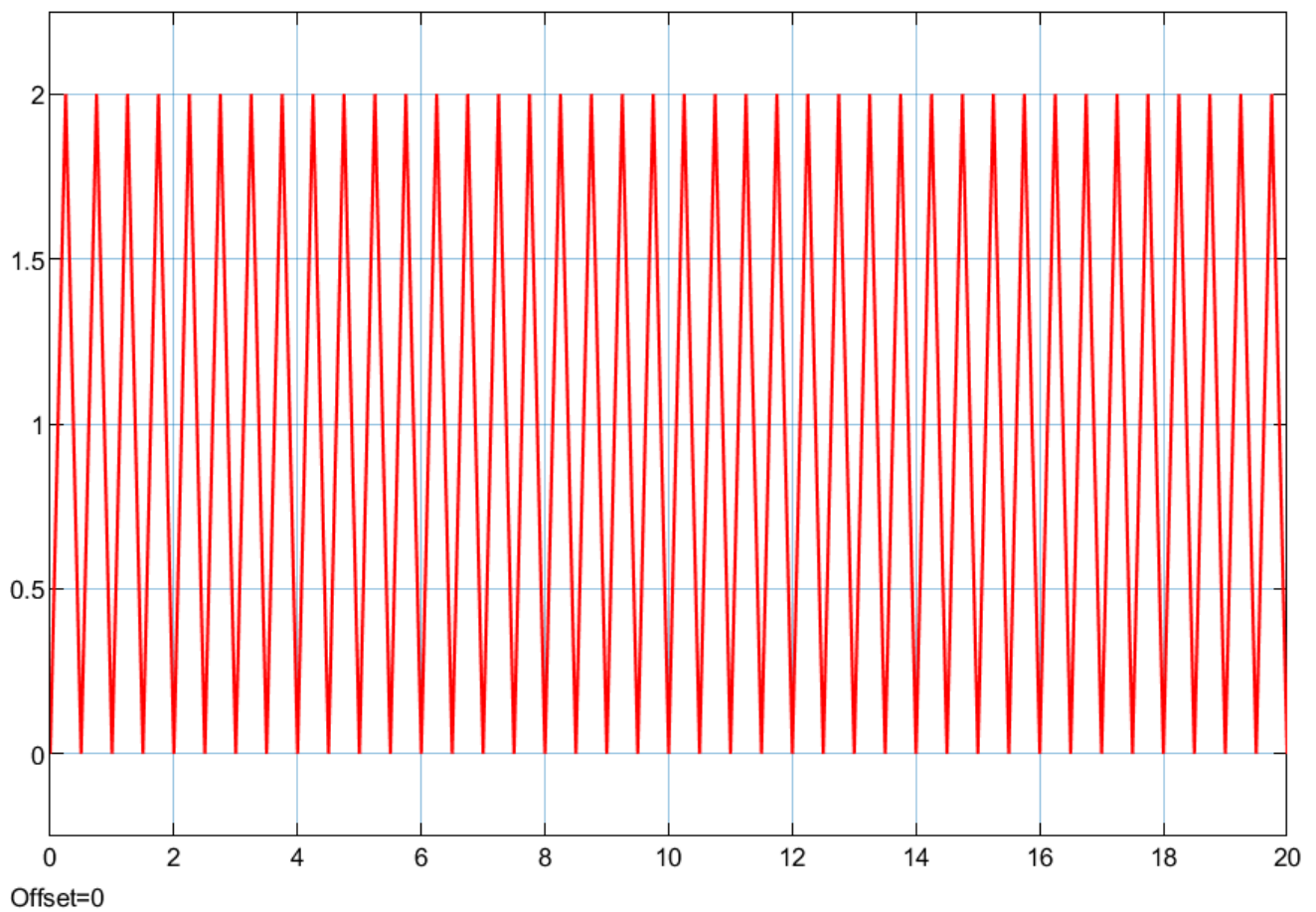


Fig. 5 – ($K_{FB} = 45$) (Unstable)

